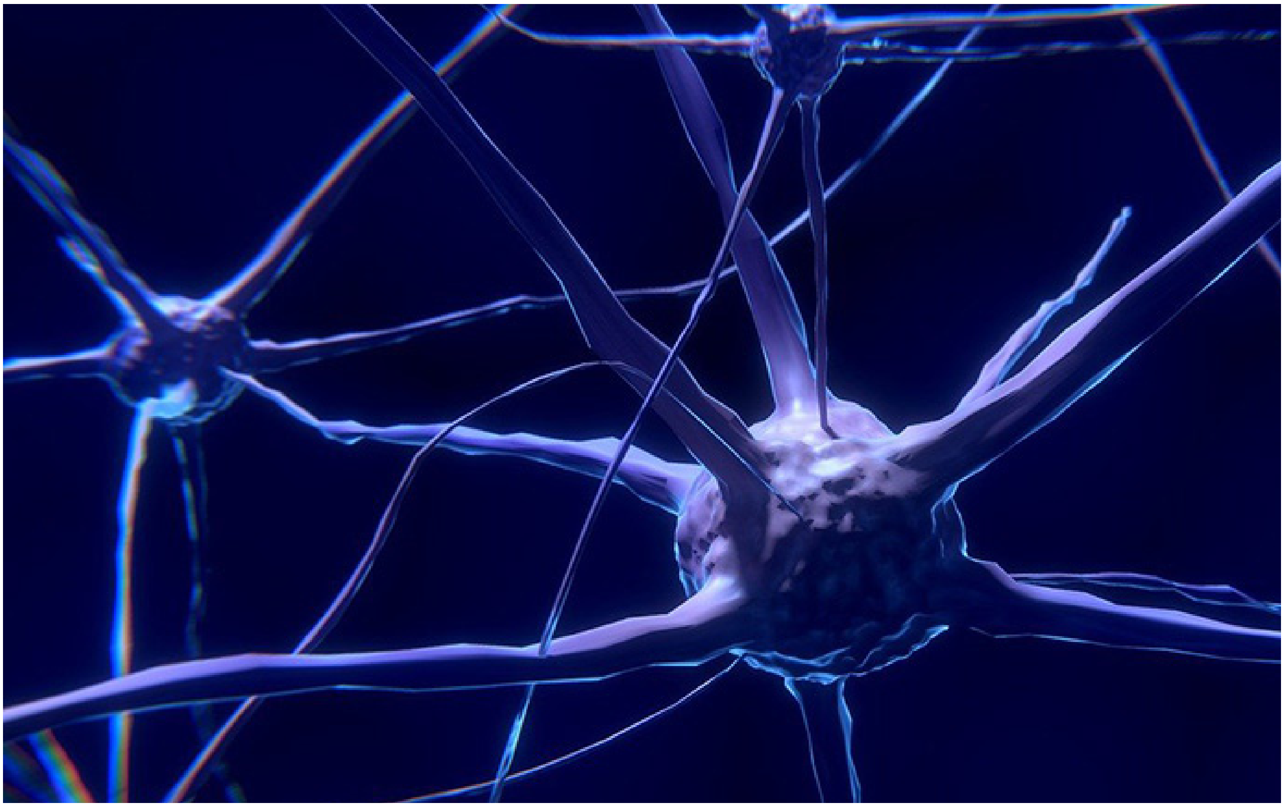

Adaptive networks

Adaptive network platforms based on AI have three important components — software control, programmable infrastructure, and analytics driven intelligence. Software control forms the basis of adaptive operations, which is supported by the automated creation and deployment of network services. These network services are deployed for scale and speed using software defined networks, network functions virtualisation and open APIs.

The programmable infrastructure is a hybrid future generation network. It comprises open, software defined network enabled physical networks and cloud-native virtual network functions. It provides advanced telemetry that delivers real-time data on the health of the network. The programmable infrastructure provides the ability to match the changing capacity needs.

Analytics driven intelligence enables intelligent automation. It can be achieved through policy management, rule engines, AI, machine learning and telemetry. It needs a robust storage repository, which records, processes and aggregates real-time and historical large scale and raw data streams. The data streams, such as log files and telemetry data, are recorded in the repository. The raw data is processed, normalised and used for advanced data models and analytics algorithms. These algorithms are used to generate actionable insights.

> Technology, through automation and artificial intelligence, is definitely one of the most disruptive sources.
>
> *— **Alain Dehaze,** CEO, The Adecco Group*

Network providers apply different kinds of machine learning techniques, which are based on the operational use cases and benefits. The techniques used are supervised learning, reinforced learning and unsupervised learning. Supervised machine learning based algorithms are trained to identify patterns such as degrading network performance. They can also be used to predict an outcome like port failure, and trigger remediation actions such as auto-adjust network bandwidth and add new capacity. This technique is commonly used and suitable for use cases in which historical data and outcomes are known. Reinforced learning involves continuous calibration of these algorithms based on previous feedback on actions. Unsupervised learning algorithms use grouping